

FITTING DISTRIBUTIONS TO GROUPED DATA USING A MAXIMUM LIKELIHOOD APPROACH

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Key words: Exponential distribution, frequency distribution; grouped data; log-logistic distribution; maximum likelihood estimation; normal distribution; Pareto distribution; Weibull distribution.

Summary: The purpose of this paper is to show how the maximum likelihood estimation procedure of Matthews and Crowther (1995) may be utilized to fit continuous distributions, such as the exponential, normal, Weibull, log-logistic and Pareto to data which appears in the form of a frequency distribution.

1. Introduction

In many instances data is only available in the form of a histogram or a frequency distribution and the aim is to fit a suitable statistical distribution to the frequency distribution. The earliest work on fitting frequency curves to data was done by Pearson (1924) between 1890 and 1900. Pearson classified the different shapes of frequency distributions into types. Elderton and Johnson (1969) discuss Pearson's various types of frequency curves in detail and present a number of applications to observed frequency distributions. Johnson and Kotz (1970) give an excellent resumé on Pearson's system of frequency curves.

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Although the problem of choosing or fitting a continuous distribution to grouped data is usually addressed in a very heuristic way, no unified procedure for obtaining maximum likelihood (ML) estimates for the parameters of the distribution is readily available, except for maximizing the likelihood function numerically. In various aspects, such as accuracy and simplicity, as well as obtaining variances and covariances of estimators, this is not a satisfactory solution to the problem.

The purpose of this paper is to show how the maximum likelihood estimation procedure of Matthews and Crowther (1995) may be implemented to fit certain continuous probability distributions to an observed frequency distribution. Estimation will entail firstly selecting a suitable continuous distribution, $F(x)$ of some random variable x , assuming that the data are represented on a continuous interval scale, and secondly estimating the parameters of the selected distribution. The method will require that the estimated cumulative probabilities, will equal the cumulative distribution curve, $F(x)$, at the upper class boundaries of the class intervals. Under this restriction, the ML estimates of the cumulative probabilities, are obtained by applying the estimation algorithm of Matthews and Crowther (1995), after which the ML estimates of the parameters characterising the cumulative distribution function are found. This paper shows how the technique may be used to fit an exponential, normal, Weibull, Pareto and log-logistic distribution to a frequency distribution.

2. Formulation

Consider n observations tabulated in a frequency distribution with k classes. The observations originate from a specified continuous distribution and

consequently information concerning the distribution is now only available in grouped format. The first and last intervals of the frequency distribution may be open ended class intervals. (See Table 1.)

Table 1. General formulation of the frequency distribution

Class Interval	Frequency
$(-\infty, x_1)$	f_1
$[x_1, x_2)$	f_2
$\vdots$	$\vdots$
$[x_{k-2}, x_{k-1})$	f_{k-1}
$[x_{k-1}, \infty)$	f_k

The frequency vector of the first $(k - 1)$ classes and corresponding vector of upper class boundaries are represented by

$$\mathbf{f} = \begin{pmatrix} f_1 \\ f_2 \\ \vdots \\ f_{k-1} \end{pmatrix} \quad \text{and} \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_{k-1} \end{pmatrix}. \quad (1)$$

It is assumed that the vector $\mathbf{f}$ is a random vector with some discrete distribution such as Poisson, multinomial or product multinomial.

Assume multinomial sampling and define $\mathbf{p}_0 = \frac{1}{n}\mathbf{f}$ as the vector of relative frequencies. Hence, the expected value and covariance matrix of $\mathbf{p}_0$ are

$$E(\mathbf{p}_0) = \boldsymbol{\pi}_0 \quad \text{and} \quad \text{Cov}(\mathbf{p}_0) = \frac{1}{n}(\text{diag}[\boldsymbol{\pi}_0] - \boldsymbol{\pi}_0\boldsymbol{\pi}_0')$$

where $\text{diag}[\boldsymbol{\pi}_0]$ is a diagonal matrix with the elements of $\boldsymbol{\pi}_0$ on the diagonal.

The vector $\boldsymbol{\pi}_0$ denotes the vector of probabilities, where the i -th element of $\boldsymbol{\pi}_0$ is the probability that an observation falls in the i -th class interval.

The vector of cumulative relative frequencies is denoted by

$$\mathbf{p} = \mathbf{C}\mathbf{p}_0 \quad (2)$$

where

$$\mathbf{C} : (k-1) \times (k-1) = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & 1 & 0 & \cdots & 0 \\ 1 & 1 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & 1 \end{pmatrix}.$$

The expected value and covariance matrix of the random vector $\mathbf{p}$ are

$$E(\mathbf{p}) = \boldsymbol{\pi} \quad (3)$$

and

$$\text{Cov}(\mathbf{p}) = \frac{1}{n} \{ \mathbf{C} \text{diag} [\mathbf{C}^{-1} \boldsymbol{\pi}] \mathbf{C}' - \boldsymbol{\pi} \boldsymbol{\pi}' \} = \mathbf{V}. \quad (4)$$

3. Estimation Procedure

By implementing the maximum likelihood estimation procedure of Matthews and Crowther (1995), it is possible to find the ML estimate of $\boldsymbol{\pi}$, under the restriction that $\boldsymbol{\pi}$ satisfies the characteristics of the specified continuous distribution. This procedure was developed for any random vector belonging to the exponential family. In the case of a frequency vector in the exponential class, this procedure is now outlined below in Proposition 1.

Proposition 1 (*ML Estimation Procedure*)

Consider a random vector of cumulative relative frequencies $\mathbf{p}$, belonging to the exponential family, with

$$E(\mathbf{p}) = \boldsymbol{\pi} \quad \text{and} \quad \text{Cov}(\mathbf{p}) = \mathbf{V}.$$

The observed $\mathbf{p}$ is the unrestricted ML estimate of $\boldsymbol{\pi}$ and the covariance matrix $\mathbf{V}$ may be a function of $\boldsymbol{\pi}$. Let $\mathbf{g}(\boldsymbol{\pi})$ be a continuous vector valued function of $\boldsymbol{\pi}$, for which the first order partial derivatives,

$$\mathbf{G}_{\pi} = \frac{\partial \mathbf{g}(\boldsymbol{\pi})}{\partial \boldsymbol{\pi}}$$

with respect to $\boldsymbol{\pi}$ exist. The ML estimate of $\boldsymbol{\pi}$, subject to the constraints $\mathbf{g}(\boldsymbol{\pi}) = \mathbf{0}$ is obtained iteratively from

$$\hat{\boldsymbol{\pi}} = \mathbf{p} - (\mathbf{G}_{\pi} \mathbf{V})' (\mathbf{G}_{\pi} \mathbf{V} \mathbf{G}_{\pi}')^* \mathbf{g}(\mathbf{p}) \quad (5)$$

where $\mathbf{G}_p = \left. \frac{\partial \mathbf{g}(\boldsymbol{\pi})}{\partial \boldsymbol{\pi}} \right|_{\boldsymbol{\pi}=\mathbf{p}}$ and $(\mathbf{G}_p \mathbf{V} \mathbf{G}'_p)^*$ is a generalized inverse of $(\mathbf{G}_p \mathbf{V} \mathbf{G}'_p)$.

The iterative procedure implies a double iteration over $\mathbf{p}$ and $\boldsymbol{\pi}$. The procedure starts with the unrestricted ML estimate of $\boldsymbol{\pi}$, as the starting value for both $\mathbf{p}$ and $\boldsymbol{\pi}$. Convergence is first obtained over $\mathbf{p}$ using (5). The converged value of $\mathbf{p}$ is then used as the next value of $\boldsymbol{\pi}$, with convergence over $\mathbf{p}$ starting again at the observed $\mathbf{p}$. In this procedure $\mathbf{V}$ is recalculated for each new value of $\boldsymbol{\pi}$ in the iterative procedure. Convergence over $\boldsymbol{\pi}$ ultimately leads to $\hat{\boldsymbol{\pi}}$, the restricted ML estimate of $\boldsymbol{\pi}$.

Proposition 2 *The asymptotic covariance matrix of $\hat{\boldsymbol{\pi}}$, under $\mathbf{g}(\boldsymbol{\pi}) = \mathbf{0}$, is*

$$\text{Cov}(\hat{\boldsymbol{\pi}}) \cong \mathbf{V} - (\mathbf{G}_\pi \mathbf{V})' (\mathbf{G}_\pi \mathbf{V} \mathbf{G}'_\pi)^* (\mathbf{G}_\pi \mathbf{V}) \quad (6)$$

which is estimated by replacing $\boldsymbol{\pi}$ by $\hat{\boldsymbol{\pi}}$.

In Matthews and Crowther (1995) it is assumed that the restrictions are linearly independent, but in Matthews and Crowther (1998), it is shown that if the restrictions are linearly dependent, it leads to the generalized inverse to be introduced in (5) and (6).

The objective is now to find the ML estimate of $\boldsymbol{\pi}$, under the constraints that the cumulative probabilities, $\boldsymbol{\pi}$, equal the cumulative distribution curve, $F(\mathbf{x})$, at the upper class boundaries $\mathbf{x}$. This implies that the ML estimate of $\boldsymbol{\pi}$ is to be obtained under the restriction

$$\boldsymbol{\pi} = F(\mathbf{x}) \quad (7)$$

which means that the vector of constraints may be formulated as

$$\mathbf{g}(\boldsymbol{\pi}) = \boldsymbol{\pi} - F(\mathbf{x}) = \mathbf{0} . \quad (8)$$

The set of constraints in Proposition 1 is essentially not unique and may be dependent. Any function say $\mathbf{g}_1(\boldsymbol{\pi})$, that implies the same constraints on $\boldsymbol{\pi}$ as $\mathbf{g}(\boldsymbol{\pi})$, may be used and will provide the same results. The objective now is to choose $\mathbf{g}(\boldsymbol{\pi})$ in such a way to simplify the calculation of derivatives and to streamline the estimation process. In some instances it is possible to find the ML estimate of $\boldsymbol{\pi}$ under constraints, by making use of traditional methods, but the procedure suggested in Proposition 1 and 2 provides an elegant and straightforward method for obtaining the ML estimates.

4. Goodness of Fit

In order to test the deviation of the observed probabilities $\mathbf{p}$, from the estimated probabilities $\hat{\boldsymbol{\pi}}$, it is convenient to formulate and test the null hypothesis

$$\mathbf{H}_0 : \mathbf{g}(\boldsymbol{\pi}) = \mathbf{0}$$

using some goodness-of-fit statistic such as the ML-statistic or the Pearson χ^2 -statistic. The Wald statistic

$$\mathbf{W} = \mathbf{g}(\mathbf{p})' (\mathbf{G}_p \mathbf{V} \mathbf{G}_p')^* \mathbf{g}(\mathbf{p}) \quad (9)$$

may be used as a measure of fit. The Wald statistic has a χ^2 -distribution with r degrees of freedom, where r is equal to the number of linear independent constraints in $\mathbf{g}(\boldsymbol{\pi})$.

Another useful measure, is the measure of discrepancy

$$\mathbf{D} = \mathbf{W}/n \quad (10)$$

which will provide more conservative results for large sample sizes. As a rule of thumb the observed and expected frequencies are considered not to deviate significantly from each other if the discrepancy is less than 0.05.

5. The Exponential Distribution

To illustrate the underlying methodology of fitting a distribution with Proposition 1, it will be shown how to fit an exponential distribution to frequency data.

The probability density function (PDF) of an exponential random variable with expected value $\mu > 0$ is

$$f(x, \mu) = \frac{1}{\mu} e^{-x/\mu}$$

with cumulative distribution function (CDF)

$$F(x, \mu) = 1 - e^{-x/\mu}$$

where $x > 0$.

To fit an exponential distribution to the frequency data in Table 1 it is required (7) that

$$\boldsymbol{\pi} = \mathbf{1} - \exp(-\theta \mathbf{x}) \quad (11)$$

where $\mathbf{1} : (k-1) \times 1$ is a vector of ones and $\mu = \theta^{-1}$. From this requirement (11) two alternative ways of performing the ML estimation procedure are described. Although the specifications of the sets of constraints seem completely different, the final results obtained are identical.

5.1 A Direct Set of Constraints

A direct set of constraints (8) may be formulated by

$$\mathbf{g}(\boldsymbol{\pi}) = \boldsymbol{\pi} - \{\mathbf{1} - \exp(-\theta \mathbf{x})\} \quad (12)$$

with matrix of partial derivatives

$$\begin{aligned} \mathbf{G}_{\boldsymbol{\pi}} &= \frac{\partial \mathbf{g}(\boldsymbol{\pi})}{\partial \boldsymbol{\pi}} \\ &= \mathbf{I} + \frac{\partial \exp(-\theta \mathbf{x})}{\partial \boldsymbol{\pi}}. \end{aligned}$$

Since

$$\theta = -\frac{\mathbf{x}' \ln(\mathbf{1} - \boldsymbol{\pi})}{\mathbf{x}' \mathbf{x}} \quad (13)$$

(see formula 11) the matrix of partial derivatives is calculated by making use of the chain rule:

$$\begin{aligned} \mathbf{G}_\pi &= \mathbf{I} + \frac{\partial \{\exp(-\theta \mathbf{x})\}}{\partial \theta} \cdot \frac{\partial \theta}{\partial \boldsymbol{\pi}} \\ &= \mathbf{I} + (\text{diag} [\exp(-\theta \mathbf{x})]) \mathbf{P} \mathbf{D} \end{aligned} \quad (14)$$

where $\mathbf{P} = \mathbf{x} (\mathbf{x}' \mathbf{x})^{-1} \mathbf{x}'$ and

$$\begin{aligned} \mathbf{D} &= \frac{\partial \ln(\mathbf{1} - \boldsymbol{\pi})}{\partial \boldsymbol{\pi}} \\ &= -(\text{diag} [\mathbf{1} - \boldsymbol{\pi}])^{-1}. \end{aligned} \quad (15)$$

The estimation procedure in Proposition 1 utilizes a double iteration over $\boldsymbol{\pi}$ and $\mathbf{p}$ starting with the observed vector of cumulative relative frequencies as the initial values for both convergencies over $\boldsymbol{\pi}$ and $\mathbf{p}$.

For each value of $\boldsymbol{\pi}$, convergence is obtained over $\mathbf{p}$ in (5) with

$$\mathbf{g}(\mathbf{p}) = \mathbf{p} - [\mathbf{1} - \exp(-\theta_p \mathbf{x})]$$

where

$$\theta_p = -\frac{\mathbf{x}' \ln(\mathbf{1} - \mathbf{p})}{\mathbf{x}' \mathbf{x}}.$$

The corresponding matrix of partial derivatives follows

$$\mathbf{G}_p = \mathbf{I} + (\text{diag} [\exp(-\theta_p \mathbf{x})]) \mathbf{P} \mathbf{D}_p$$

where

$$\mathbf{D}_p = -(\text{diag} [\mathbf{1} - \mathbf{p}])^{-1}.$$

Once convergence over $\mathbf{p}$ is attained, the converged value of $\mathbf{p}$ is used as the next value of $\boldsymbol{\pi}$, with covariance matrix

$$\mathbf{V} = \frac{1}{n} \{ \mathbf{C} (\text{diag} [\mathbf{C}^{-1} \boldsymbol{\pi}]) \mathbf{C}' - \boldsymbol{\pi} \boldsymbol{\pi}' \}.$$

The matrix of partial derivatives is

$$\mathbf{G}_\pi = \mathbf{I} + (\text{diag}[\exp(-\theta_\pi \mathbf{x})]) \mathbf{P} \mathbf{D}_\pi$$

where

$$\theta_\pi = -\frac{\mathbf{x}' \ln(\mathbf{1} - \boldsymbol{\pi})}{\mathbf{x}' \mathbf{x}} \quad \text{and} \quad \mathbf{D}_\pi = -(\text{diag}[\mathbf{1} - \boldsymbol{\pi}])^{-1}$$

are the corresponding values for convergence over $\boldsymbol{\pi}$.

Convergence over $\boldsymbol{\pi}$ ultimately leads to the restricted ML estimate of $\boldsymbol{\pi}$ namely $\hat{\boldsymbol{\pi}}$ and hence the ML estimate of μ is

$$\hat{\mu} = \frac{1}{\hat{\theta}} = -\left(\frac{\mathbf{x}' \ln(\mathbf{1} - \hat{\boldsymbol{\pi}})}{\mathbf{x}' \mathbf{x}}\right)^{-1}.$$

This iterative process is illustrated in Example 1.

Example 1

A total of $n = 100$ observations are simulated from an exponential distribution with expected value $\mu = \theta^{-1} = 50$. The grouped data set is shown in Table 2.

Table 2. Sample of 100 observations simulated from an exponential distribution

Class interval	Frequency
[0, 12.5)	17
[12.5, 25)	14
[25, 50)	31
[50, 100)	26
[100, ∞)	12

Table 3 shows the various values of $\boldsymbol{\pi}$ and $\mathbf{p}$ in the double iteration process, with corresponding values for μ . The results in Table 3 can be calculated directly, or can be obtained using the SAS program *EXPI.SAS* listed in the Appendix.

Table 3. Double iteration process

Iteration over π		Iteration over $\mathbf{p}$			
		$j = 1$	$j = 2$	$j = 3$	$j = 4$
$i = 1$	$\begin{pmatrix} 0.1700 \\ 0.3100 \\ 0.6200 \\ 0.8800 \end{pmatrix}$ $\mu_\pi = 48.83$	$\begin{pmatrix} 0.1700 \\ 0.3100 \\ 0.6200 \\ 0.8800 \end{pmatrix}$ $\mu_p = 48.83$	$\begin{pmatrix} 0.2373 \\ 0.4184 \\ 0.6620 \\ 0.8862 \end{pmatrix}$ $\mu_p = 46.03$	$\begin{pmatrix} 0.2380 \\ 0.4194 \\ 0.6629 \\ 0.8863 \end{pmatrix}$ $\mu_p = 45.99$	$\begin{pmatrix} 0.2380 \\ 0.4194 \\ 0.6629 \\ 0.8863 \end{pmatrix}$ $\mu_p = 45.99$
$i = 2$	$\begin{pmatrix} 0.2380 \\ 0.4194 \\ 0.6629 \\ 0.8863 \end{pmatrix}$ $\mu_\pi = 45.99$	$\begin{pmatrix} 0.1700 \\ 0.3100 \\ 0.6200 \\ 0.8800 \end{pmatrix}$ $\mu_p = 48.83$	$\begin{pmatrix} 0.2137 \\ 0.3820 \\ 0.6186 \\ 0.8563 \end{pmatrix}$ $\mu_p = 51.63$	$\begin{pmatrix} 0.2147 \\ 0.3833 \\ 0.6197 \\ 0.8553 \end{pmatrix}$ $\mu_p = 51.72$	$\begin{pmatrix} 0.2147 \\ 0.3833 \\ 0.6197 \\ 0.8553 \end{pmatrix}$ $\mu_p = 51.72$
$i = 3$	$\begin{pmatrix} 0.2147 \\ 0.3833 \\ 0.6197 \\ 0.8553 \end{pmatrix}$ $\mu_\pi = 51.72$	$\begin{pmatrix} 0.1700 \\ 0.3100 \\ 0.6200 \\ 0.8800 \end{pmatrix}$ $\mu_p = 48.83$	$\begin{pmatrix} 0.2143 \\ 0.3829 \\ 0.6196 \\ 0.8570 \end{pmatrix}$ $\mu_p = 51.49$	$\begin{pmatrix} 0.2152 \\ 0.3841 \\ 0.6207 \\ 0.8561 \end{pmatrix}$ $\mu_p = 51.57$	$\begin{pmatrix} 0.2152 \\ 0.3841 \\ 0.6207 \\ 0.8561 \end{pmatrix}$ $\mu_p = 51.57$
$i = 4$	$\begin{pmatrix} 0.2152 \\ 0.3841 \\ 0.6207 \\ 0.8561 \end{pmatrix}$ $\mu_\pi = 51.57$	$\begin{pmatrix} 0.1700 \\ 0.3100 \\ 0.6200 \\ 0.8800 \end{pmatrix}$ $\mu_p = 48.83$	$\begin{pmatrix} 0.2143 \\ 0.3828 \\ 0.6196 \\ 0.8570 \end{pmatrix}$ $\mu_p = 51.49$	$\begin{pmatrix} 0.2152 \\ 0.3841 \\ 0.6207 \\ 0.8561 \end{pmatrix}$ $\mu_p = 51.58$	$\begin{pmatrix} 0.2152 \\ 0.3841 \\ 0.6207 \\ 0.8561 \end{pmatrix}$ $\mu_p = 51.58$
$i = 5$	$\begin{pmatrix} 0.2152 \\ 0.3841 \\ 0.6207 \\ 0.8561 \end{pmatrix}$ $\mu_\pi = 51.58$	$\begin{pmatrix} 0.1700 \\ 0.3100 \\ 0.6200 \\ 0.8800 \end{pmatrix}$ $\mu_p = 48.83$	$\begin{pmatrix} 0.2143 \\ 0.3828 \\ 0.6196 \\ 0.8570 \end{pmatrix}$ $\mu_p = 51.49$	$\begin{pmatrix} 0.2152 \\ 0.3841 \\ 0.6207 \\ 0.8561 \end{pmatrix}$ $\mu_p = 51.58$	$\begin{pmatrix} 0.2152 \\ 0.3841 \\ 0.6207 \\ 0.8561 \end{pmatrix}$ $\mu_p = 51.58$

The procedure starts with the unrestricted ML estimate of π

$$\pi = \mathbf{p} = \begin{pmatrix} 0.1700 \\ 0.3100 \\ 0.6200 \\ 0.8800 \end{pmatrix}$$

(the observed vector of cumulative relative frequencies) and after convergence the restricted ML estimate of π

$$\hat{\pi} = \begin{pmatrix} 0.2152 \\ 0.3841 \\ 0.6207 \\ 0.8561 \end{pmatrix}$$

is obtained. (See Table 3.) The elements of $\hat{\pi}$ now follow a cumulative exponential curve on the upper class boundaries and hence the ML estimate

$$\hat{\mu} = - \left(\frac{\mathbf{x}' \ln(\mathbf{1} - \hat{\pi})}{\mathbf{x}' \mathbf{x}} \right)^{-1} = 51.58$$

follows. The estimated exponential distribution

$$f(x) = \frac{1}{51.58} \exp\left(-\frac{x}{51.58}\right) \quad (16)$$

is shown in Figure 1, together with the observed frequency distribution (solid line) and estimated frequency distribution (dashed line).

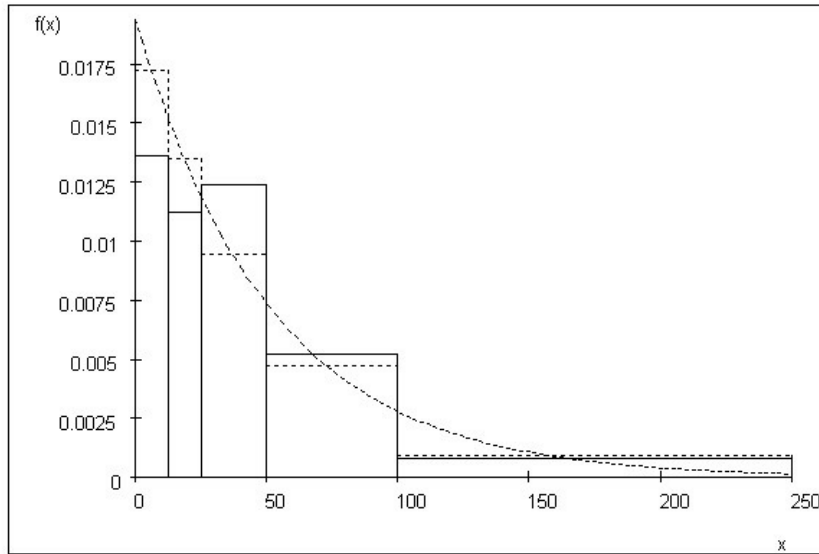


Figure 1. The estimated exponential distribution with the observed and estimated frequency distribution

5.2 Constraints in terms of a Linear Model

An alternative formulation of the vector of constraints may be developed. Since (11) implies that

$$\ln(\mathbf{1} - \boldsymbol{\pi}) = -\theta \mathbf{x} \quad (17)$$

this constitutes a linear model. In any practical situation this means that the plot of $\ln(\mathbf{1} - \boldsymbol{\pi})$ against $\mathbf{x}$ is an approximate straight line. Therefore, the vector of constraints may now be expressed in terms of a new $\mathbf{g}(\boldsymbol{\pi})$ namely

$$\mathbf{g}(\boldsymbol{\pi}) = \mathbf{Q} \ln(\mathbf{1} - \boldsymbol{\pi}) = \mathbf{0} \quad (18)$$

where $\mathbf{Q} = \mathbf{I} - \mathbf{x}(\mathbf{x}'\mathbf{x})^{-1}\mathbf{x}'$ is the projection matrix orthogonal to $\mathbf{x}$.

Since the $\text{rank}(\mathbf{Q}) = (k - 1) - 1$, the vector of constraints, $\mathbf{g}(\boldsymbol{\pi})$, consists of $k - 2$ linear independent functions. The matrix of partial derivatives is now much simpler than in the previous formulation (14) with

$$\mathbf{G}_\pi = \frac{\partial \mathbf{g}(\boldsymbol{\pi})}{\partial \boldsymbol{\pi}} = \mathbf{Q} \frac{\partial \ln(\mathbf{1} - \boldsymbol{\pi})}{\partial \boldsymbol{\pi}} = \mathbf{Q}\mathbf{D} \quad (19)$$

where $\mathbf{D} = -(\text{diag}[\mathbf{1} - \boldsymbol{\pi}])^{-1}$.

After obtaining $\hat{\boldsymbol{\pi}}$, the MLE of $\boldsymbol{\pi}$ under (18) with the ML estimation procedure, the ML estimates of θ and μ are

$$\hat{\theta} = -\frac{\mathbf{x}' \ln(\mathbf{1} - \hat{\boldsymbol{\pi}})}{\mathbf{x}'\mathbf{x}}$$

and

$$\hat{\mu} = \frac{1}{\hat{\theta}}.$$

Using the multivariate delta theorem

$$\begin{aligned} \text{Var}(\hat{\theta}) &\cong \left\{ \frac{\partial \theta}{\partial \boldsymbol{\pi}} \right\} \text{Cov}(\hat{\boldsymbol{\pi}}) \left\{ \frac{\partial \theta}{\partial \boldsymbol{\pi}} \right\}' \\ &= \left\{ \frac{\mathbf{x}'}{\mathbf{x}'\mathbf{x}} \mathbf{D} \right\} \text{Cov}(\hat{\boldsymbol{\pi}}) \left\{ \frac{\mathbf{x}'}{\mathbf{x}'\mathbf{x}} \mathbf{D} \right\}' \end{aligned}$$

where $\text{Cov}(\hat{\boldsymbol{\pi}})$ is given in Proposition 2.

Applying the multivariate delta theorem again it follows that

$$\begin{aligned}\text{Var}(\hat{\mu}) &\cong \left\{ \frac{\partial \mu}{\partial \theta} \right\}^2 \text{Var}(\hat{\theta}) \\ &= \frac{1}{\theta^4} \text{Var}(\hat{\theta}).\end{aligned}$$

Hence

$$\hat{\mu} \cong N\left(\mu, \frac{1}{\theta^4} \text{Var}(\hat{\theta})\right).$$

Example 2

In this example the estimation of the exponential distribution to the simulated frequency distribution in Table 2 is revisited. The vector of constraints (18) is now formulated in terms of a linear model.

The ML estimates are tabulated in Table 4. It follows that the ML estimation procedure with the vector of constraints in (18) yields exactly the same results as in the previous formulation of (12), although the intermediate iterations differ. Other relevant statistics are also calculated and summarised in Table 4.

Table 4. Results for the estimated exponential distribution

MLE		Wald			Discrepancy
Estimate	Std. error	Statistic	df	prob	
$\hat{\mu} = 51.578$	$\hat{\sigma}_{\hat{\mu}} = 5.654$	$W = 4.295$	3	0.2313	$D = 0.043$

As can be expected, the Wald statistic and the measure of discrepancy indicate an adequate fit ($D < 0.05$). A 95% confidence interval for μ follows from $\hat{\mu} \pm 1.96\hat{\sigma}_{\hat{\mu}}$, i.e. (40.496, 62.660).

6. Other Distributions

In addition to the exponential distribution, explained in detail in the previous section, it will be shown how to fit the normal, Weibull, log-logistic and the

Pareto distributions to grouped data. These distributions are all summarised in Table 5.

Estimation in this section will be done by constructing the vector of constraints in terms of a linear model. This method is preferred due to the simplicity and the overall generalization of the technique. First the restricted ML estimate of π will be estimated, which will then in turn lead to the estimation of the ML estimates of the corresponding parameters of the distributions.

Table 5. Characteristics of distributions considered

	PDF and CDF	Mean and Variance
Exponential	$f(x, \mu) = \frac{1}{\mu} e^{-x/\mu}$ $F(x, \mu) = 1 - e^{-x/\mu}$	μ $\sigma^2 = \mu^2$
Normal	$f(x, \mu, \sigma^2) = \phi\left(\frac{x - \mu}{\sigma}\right)$ $F(x, \mu, \sigma^2) = \Phi\left(\frac{x - \mu}{\sigma}\right)$	μ σ^2
Weibull	$f(x, \kappa, \theta) = \frac{\kappa}{\theta^\kappa} x^{\kappa-1} \exp\left[-\left(\frac{x}{\theta}\right)^\kappa\right]$ $F(x, \kappa, \theta) = 1 - \exp\left[-\left(\frac{x}{\theta}\right)^\kappa\right]$	$\mu = \theta \left[\Gamma\left(1 + \frac{1}{\kappa}\right)\right]$ $\sigma^2 = \theta^2 \left[\Gamma\left(1 + \frac{2}{\kappa}\right) - \Gamma^2\left(1 + \frac{1}{\kappa}\right)\right]$
Log-logistic	$f(x, \kappa, \theta) = \frac{e^\theta \kappa x^{\kappa-1}}{(1 + e^\theta x^\kappa)^2}$ $F(x, \kappa, \theta) = \frac{e^\theta x^\kappa}{1 + e^\theta x^\kappa}$	$\mu = \exp\left(-\frac{\theta}{\kappa}\right) \left[\Gamma\left(1 + \frac{1}{\kappa}\right) \Gamma\left(1 - \frac{1}{\kappa}\right)\right]$ $\sigma^2 = \exp\left(-\frac{2\theta}{\kappa}\right) \left[\Gamma\left(1 + \frac{2}{\kappa}\right) \Gamma\left(1 - \frac{2}{\kappa}\right) - \Gamma^2\left(1 + \frac{1}{\kappa}\right) \Gamma^2\left(1 - \frac{1}{\kappa}\right)\right]$
Pareto	$f(x, \kappa, \theta) = \kappa \theta^\kappa x^{-(\kappa+1)}$ $F(x, \kappa, \theta) = 1 - \left(\frac{x}{\theta}\right)^{-\kappa}$	$\mu = \frac{\kappa \theta}{\kappa - 1}$ $\sigma^2 = \frac{\kappa \theta^2}{(\kappa - 1)^2 (\kappa - 2)}$

In the case of the distributions $F(x, \beta)$, specified in Table 5, the requirement

$$F(\mathbf{x}, \beta) = \pi \quad (20)$$

where $F(\mathbf{x}, \beta)$ denotes the distribution function at the upper class boundaries $\mathbf{x}$ with parameter vector β , may be transformed into the linear model

$$\mathbf{h}(\pi) = \mathbf{X}\alpha \quad (21)$$

which implies that the estimation procedure may be performed in the three steps outlined below.

Step 1: The vector of constraints is given by

$$\mathbf{g}(\pi) = \mathbf{Q}\mathbf{h}(\pi) = \mathbf{0}$$

with matrix of partial derivatives

$$\mathbf{G}_\pi = \mathbf{Q}\mathbf{D}$$

where $\mathbf{Q} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ and $\mathbf{D} = \frac{\partial \mathbf{h}(\pi)}{\partial \pi}$.

Step 2: The ML estimate of α follows as

$$\hat{\alpha} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{h}(\hat{\pi})$$

with covariance matrix

$$\text{Cov}(\hat{\alpha}) = \left\{ (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{D} \right\} \text{Cov}(\hat{\pi}) \left\{ (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{D} \right\}' .$$

Step 3: The ML estimates, of the original parameters, $\hat{\beta}$, are obtained from $\hat{\alpha}$ with

$$\text{Cov}(\hat{\beta}) = \mathbf{B} \text{Cov}(\hat{\alpha}) \mathbf{B}'$$

where $\mathbf{B} = \frac{\partial \beta}{\partial \alpha}$. From the multivariate delta theorem, it follows that

$$\hat{\beta} \cong N(\beta, \mathbf{B} \text{Cov}(\hat{\alpha}) \mathbf{B}') .$$

To fit the various continuous distributions in Table 5 to grouped data by means of the three steps listed above, a summary of the constraints and derivatives are given in Table 6(A) and Table 6(B).

Table 6(A). Constraints

	β	$\mathbf{h}(\boldsymbol{\pi}) = \mathbf{X}\boldsymbol{\alpha}$		
		$\mathbf{h}(\boldsymbol{\pi})$	$\mathbf{X}$	$\boldsymbol{\alpha}$
Exponential	$\mu = \frac{1}{\alpha}$	$\ln(\mathbf{1} - \boldsymbol{\pi})$	$(-\mathbf{x})$	$\frac{1}{\mu}$
Normal	$\begin{pmatrix} \mu \\ \sigma \end{pmatrix} = \begin{pmatrix} \frac{\alpha_2}{\alpha_1} \\ \frac{1}{\alpha_1} \end{pmatrix}$	$\Phi^{-1}(\boldsymbol{\pi})$	$(\mathbf{x} \quad -\mathbf{1})$	$\begin{pmatrix} \frac{1}{\sigma} \\ \frac{\mu}{\sigma} \end{pmatrix}$
Weibull	$\begin{pmatrix} \kappa \\ \theta \end{pmatrix} = \begin{pmatrix} \alpha_1 \\ e^{\frac{\alpha_2}{\alpha_1}} \end{pmatrix}$	$\ln[-\ln(\mathbf{1} - \boldsymbol{\pi})]$	$(\ln \mathbf{x} \quad -\mathbf{1})$	$\begin{pmatrix} \kappa \\ \kappa \ln \theta \end{pmatrix}$
Log-logistic	$\begin{pmatrix} \kappa \\ \theta \end{pmatrix} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$	$\ln\left(\frac{\boldsymbol{\pi}}{\mathbf{1} - \boldsymbol{\pi}}\right)$	$(\ln \mathbf{x} \quad \mathbf{1})$	$\begin{pmatrix} \kappa \\ \theta \end{pmatrix}$
Pareto	$\begin{pmatrix} \kappa \\ \theta \end{pmatrix} = \begin{pmatrix} \alpha_1 \\ e^{\frac{\alpha_2}{\alpha_1}} \end{pmatrix}$	$\ln(\mathbf{1} - \boldsymbol{\pi})$	$(-\ln \mathbf{x} \quad \mathbf{1})$	$\begin{pmatrix} \kappa \\ \kappa \ln \theta \end{pmatrix}$

Table 6(B). Derivatives

	$\mathbf{D} = \frac{\partial \mathbf{h}(\boldsymbol{\pi})}{\partial \boldsymbol{\pi}}$	$\mathbf{B} = \frac{\partial \boldsymbol{\beta}}{\partial \boldsymbol{\alpha}}$
Exponential	$-(\text{diag} [\mathbf{1}-\boldsymbol{\pi}])^{-1}$	$-\frac{1}{\alpha^2}$
Normal	$(\text{diag} [\phi(\Phi^{-1}(\boldsymbol{\pi}))])^{-1}$	$\begin{pmatrix} -\frac{\alpha_2}{\alpha_1^2} & \frac{1}{\alpha_1} \\ -\frac{1}{\alpha_1^2} & 0 \end{pmatrix}$
Weibull	$-(\text{diag} [\ln(\mathbf{1}-\boldsymbol{\pi})])^{-1} (\text{diag} [\mathbf{1}-\boldsymbol{\pi}])^{-1}$	$\begin{pmatrix} 1 & 0 \\ -\frac{\alpha_2}{\alpha_1^2} \cdot e^{\frac{\alpha_2}{\alpha_1}} & \frac{1}{\alpha_1} \cdot e^{\frac{\alpha_2}{\alpha_1}} \end{pmatrix}$
Log-logistic	$(\text{diag} [\boldsymbol{\pi}])^{-1} + (\text{diag} [\mathbf{1}-\boldsymbol{\pi}])^{-1}$	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
Pareto	$-(\text{diag} [\mathbf{1}-\boldsymbol{\pi}])^{-1}$	$\begin{pmatrix} 1 & 0 \\ -\frac{\alpha_2}{\alpha_1^2} \cdot e^{\frac{\alpha_2}{\alpha_1}} & \frac{1}{\alpha_1} \cdot e^{\frac{\alpha_2}{\alpha_1}} \end{pmatrix}$

Example 3

The annual income (in R1000) of a group of insurance policy holders is reported in Table 7.

Table 7. Income of a Group of Insurance Policy Holders

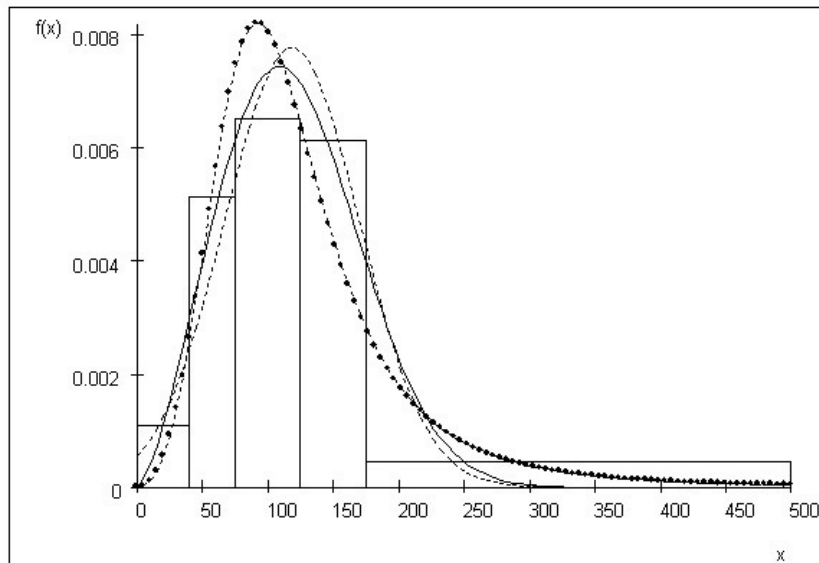
Income (in R1000)	[0, 40)	[40, 75)	[75, 125)	[125, 175)	[175, ∞)
Frequency	9	37	67	63	30

For this example the normal, Weibull and log-logistic distributions are fitted and the results are given in Table 8.

Table 8. Estimates of Parameters and Test Statistics

	MLE					Wald			Discrep- ancy
	$\hat{\beta}$	Estimate	Std. Error	$\hat{\mu}$	$\hat{\sigma}$	Statistic	df	prob	
Normal	$\hat{\mu}$ $\hat{\sigma}$	118.4 51.4	3.7604 3.0834	118.4 51.4		3.980	2	0.1367	0.019
Weibull	$\hat{\kappa}$ $\hat{\theta}$	2.4647 134.44	0.1675 4.2552	119.2 51.7		1.293	2	0.5240	0.006
Log-logistic	$\hat{\kappa}$ $\hat{\theta}$	3.3337 -15.710	0.2293 1.0883	129.7 88.0		8.731	2	0.0127	0.042

According to the Wald statistic the Weibull distribution provided the best fit, followed by the normal distribution. The distributions are illustrated in Figure 2. In constructing the histogram, it is assumed that the income of all the policy holders in the sample is less than R500 000.

**Figure 2.** Income distribution of policy holders

Normal: <i>Dashed</i>	Weibull: <i>Solid</i>	Log-logistic: <i>Dots</i>
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7. Simulation Studies

To demonstrate the accuracy of the procedure, two simulation studies were performed, the first where the exponential distribution was fitted and the second where the normal distribution was fitted.

7.1 Exponential Distribution

In this study 1000 samples were simulated from an exponential distribution with expected value $\mu = 50$. Each sample consisted of 100 observations and were classified into the 5 class intervals of Table 2. The ML estimate for μ as well as its estimated standard error was calculated for each of the 1000 generated frequency distributions. The true theoretical values as well as the descriptive statistics for the ML estimates are summarised in Table 9.

Table 9. Simulation results for the exponential distribution

Function	Theoretical Value	Mean	Std. deviation	P_5	Median	P_{95}
μ	50	50.127	5.727	41.381	49.676	59.956
$\sigma_{\hat{\mu}}$	5.458	5.487	0.716	4.421	5.418	6.732

Since the data set was simulated from an exponential distribution with expected value $\mu = 50$ the true population value for π follows from

$$\pi = \mathbf{1} - \exp\left(-\frac{\mathbf{x}}{50}\right) = \begin{pmatrix} 0.2212 \\ 0.3935 \\ 0.6321 \\ 0.8647 \end{pmatrix} \quad (22)$$

which implies that the standard error for $\hat{\mu}$ is

$$\begin{aligned} \sigma_{\hat{\mu}} &\cong \sqrt{50^4 \text{Var}(\hat{\theta})} \\ &= 5.458. \end{aligned} \quad (23)$$

(See Table 9.) This compares well with the standard deviation of 5 of the mean of an ungrouped sample of 100 observations from this exponential distribution.

From Table 9 it follows that the mean and the median of the ML estimates are relatively close to the theoretical values. Further it is known that approximately 90% of the $\hat{\mu}$ -values are within 1.645 standard deviations from $\mu = 50$, i.e. $1.645\sigma_{\hat{\mu}} = 8.978$. This is in accordance with the fifth and the ninety-fifth percentile of the $\hat{\mu}$ -values tabulated in Table 9. The standard deviation of the $\hat{\mu}$ -values is also quite close to the standard error $\sigma_{\hat{\mu}}$.

7.2 Normal Distribution

Similar to the previous simulation study, 1000 samples were simulated, each sample containing 100 observations. These samples were all simulated from a normal population with mean $\mu = 58$ and standard deviation $\sigma = 15$ and classified into the following class intervals:

Class Intervals				
$(-\infty, 40)$	$[40, 50)$	$[50, 60)$	$[60, 75)$	$[75, \infty)$

The ML estimates for μ and σ were then calculated with corresponding standard errors. The results are tabulated in Table 10.

Table 10. Simulation results for the normal distribution

Function	Theoretical Value	Mean	Std. deviation	P_5	Median	P_{95}
μ	58.000	57.993	1.548	55.512	57.945	60.598
$\sigma_{\hat{\mu}}$	1.569	1.562	0.146	1.343	1.550	1.826
σ	15.000	14.915	1.384	12.797	14.823	17.376
$\sigma_{\hat{\sigma}}$	1.341	1.340	0.171	1.091	1.320	1.653

As is evident from Table 10 the mean and the median of each of the ML estimates compare extremely well with the theoretical true values (approximate

in the case of $\sigma_{\hat{\mu}}$ and $\sigma_{\hat{\sigma}}$). It is also interesting to note that standard deviations for $\hat{\mu}$ and $\hat{\sigma}$ are close to the standard errors $\sigma_{\hat{\mu}}$ and $\sigma_{\hat{\sigma}}$. To evaluate the fifth and the ninety-fifth percentiles the margin of error for the 90% confidence intervals are summarised in Table 11.

Table 11. Margin of Error for the ML estimates of the normal distribution

Estimate	Std. Error	Margin of Error
$\hat{\mu}$	$\sigma_{\hat{\mu}}$	$1.645\sigma_{\hat{\mu}} = 2.581$
$\hat{\sigma}$	$\sigma_{\hat{\sigma}}$	$1.645\sigma_{\hat{\sigma}} = 2.206$

It is known that approximately 90% of the $\hat{\mu}$ -values should be in the interval (55.419, 60.581), while 90% of the $\hat{\sigma}$ -values should be in the interval (12.794, 17.206). This compares well with the simulated values in Table 10.

8. Conclusion

A general methodology is presented for obtaining ML estimates of parameters of continuous distributions, in the case of grouped data. Apart from numerically maximizing likelihood functions, no such procedure is known in literature.

It is shown how the estimation of these parameters as well as statistical inference relating to these parameters can be done. An iterative procedure is used for obtaining ML estimates of expected frequencies under restrictions and convergence to any degree of accuracy is extremely fast. The simulation studies demonstrate the accuracy of the estimation procedure.

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9. Appendix

EXP1.SAS

```

proc iml worksizes= 60;
f={17,14,31,26,12}; n=f[+];
x={12.5,25,50,100};
C={1 0 0 0,
    1 1 0 0,
    1 1 1 0,
    1 1 1 1};
k=nrow(f); k1=k-1;
v1=J(k1,1,1);
Px=x*inv(x`*x)*x`;
p=C*f[1:k1]/n;
i=0; p0=p; diff1=1;
do while (diff1 > 1e-9);
    i=i+1; pi=p; p=p0;
    V=(C*diag(inv(C)*pi)*C`-pi*pi`)/n;
    thetapi=-(x`*log(v1-pi))/(x`*x); mupi=1/thetapi;
    Dpi=-inv(diag(v1-pi));
    Gpi=I(k1)+diag(exp(-thetapi*x))*Px*Dpi;
    j=0; diff=1;
    do while (diff > 1e-9);
        j=j+1; pv=p;
        thetap=-(x`*log(v1-p))/(x`*x); mup=1/thetap;
        Dp=-inv(diag(v1-p));

```

```

Gp=I(kl)+diag(exp(-thetap*x))*Px*Dp;
g=p-(v1-exp(-thetap*x));
print i j g pi p thetapi mupi thetap mup;
p=p-(Gpi*V)\'*ginv(Gp*V*Gpi\')*g;
diff=sqrt((p-pv)\'*(p-pv));
end;
diff1=sqrt((p-pi)\'*(p-pi));
end;

```